

Brucella melitensis Brucellosis:

Comprehensive Disease Characterization

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Summary

Brucella melitensis brucellosis is the most common and most clinically severe form of human brucellosis, a globally distributed bacterial zoonosis acquired primarily from small ruminants (sheep and goats). The etiologic agent, *Brucella melitensis*, is a Gram-negative, facultative intracellular coccobacillus transmitted to humans chiefly through ingestion of unpasteurized dairy products, and secondarily via inhalation and direct skin/mucosal contact. It is fundamentally an **infectious, non-heritable disease**; the only "genetic" dimension is host susceptibility, which is modulated by polymorphisms in cytokine and innate-immune-receptor genes (IL4, IL18, TNF, IL10, TGFB1, TLR4). Because the pathogen resides and replicates inside host macrophages within an endoplasmic-reticulum-derived vacuole, it produces a **chronic, relapsing, multisystem febrile illness** requiring prolonged combination antibiotic therapy.

Mechanistically, the disease follows a well-characterized causal chain from **oral/mucosal entry → uptake by macrophages and dendritic cells → VirB Type IV secretion system (T4SS)-driven remodeling of the Brucella-containing vacuole (BCV) into a replicative ER-derived organelle → intracellular replication and immune subversion (Omp25-mediated TNF-α suppression, inhibition of macrophage apoptosis) → hematogenous dissemination to the reticuloendothelial system, bone/joints, liver, spleen, genitourinary tract, heart and CNS → granulomatous inflammation and the clinical syndrome of undulant fever, arthralgia and hepatosplenomegaly**. Host control depends on a Th1/IFN-γ/IL-12 cellular immune response; when this fails, the organism persists in a myeloid reservoir, producing chronic and focal disease.

Clinically, the three most frequent manifestations are **fever (~83%), arthralgia/arthritis (~67%), and hepatosplenomegaly (~61%)**. Osteoarticular disease (spondylitis, sacroiliitis, osteomyelitis) is the most common complication; neurobrucellosis is rare (~0.9%) and

endocarditis, though rare, is the leading cause of brucellosis death. Diagnosis rests on serology (Rose Bengal, standard agglutination test, ELISA) plus blood/tissue culture, with PCR adding sensitivity. First-line treatment is **doxycycline plus rifampicin or doxycycline plus an aminoglycoside** for a prolonged course; monotherapy relapses. Overall mortality is low. With no licensed human vaccine, prevention is a **One-Health enterprise** centered on livestock vaccination (*B. melitensis* Rev1), pasteurization, and reservoir control.

1. Disease Information

Overview. *Brucella melitensis* brucellosis is a systemic zoonotic infection and the predominant cause of human brucellosis worldwide. It is an aggregated, disease-level entity (the knowledge here is drawn from case series, cohorts, registries, and mechanistic studies rather than a single-patient EHR source). The disease is characterized by a variable clinical presentation ranging from acute febrile illness to chronic relapsing multisystem disease.

Key identifiers. - **MONDO:** MONDO:0001972 - **MeSH:** Brucellosis; *Brucella melitensis* (organism) - **ICD-10:** A23.0 (Brucellosis due to *Brucella melitensis*); **ICD-11:** 1B95 - **NCBI Taxonomy (organism):** *Brucella melitensis*, txid29459

Synonyms/alternative names. Malta fever, Mediterranean fever, undulant fever, goat/sheep brucellosis, febris melitensis. (Bang's disease historically refers to *B. abortus*.)

"Brucella melitensis, one of the organisms responsible for causing the disease in sheep and goat, is responsible for the disease in humans. The disease is transmitted mainly from sheep and goat to humans via ingestion (typically through milk), inhalation, abrasion and so on."

— PMID: 41132010

2. Etiology

Primary cause — infectious. The disease is caused entirely by infection with *Brucella melitensis*, a facultative intracellular Gram-negative coccobacillus. Of the four *Brucella* species significant to human health (*B. melitensis*, *B. abortus*, *B. suis*, *B. canis*), *B. melitensis* is the most virulent and the leading cause of human disease. Small ruminants are the reservoir: a

systematic review/meta-analysis of 2010–2023 data found pooled *B. melitensis* prevalence of **8.07% (95% CI 6.36–9.78%) in sheep, 2.46% (95% CI 1.70–3.21%) in goats, and 5.54% combined (95% CI 4.63–6.45%)** (PMID: 41132010).

Risk factors (environmental/behavioral — the dominant determinants). - **Dietary:** consumption of unpasteurized/fresh milk and dairy or undercooked meat (aOR 5.70, 95% CI 1.94–16.76) (PMID: 41860883). - **Occupational/contact:** rearing livestock at home (aOR 3.75, 95% CI 1.36–10.32), frequent contact with animal manure (aOR 3.29, 95% CI 1.29–8.47); farmers, shepherds, veterinarians, abattoir and laboratory workers are high-risk. - **Socioeconomic:** lack of formal education (aOR 4.05, 95% CI 1.02–16.01) (PMID: 41860883). - **Demographic:** male sex and adult working age (males 71% of cases in a large Chinese series, 2.45× females) (PMID: 41632803).

Genetic risk factors (host susceptibility, not causal). Cytokine and innate-immune-receptor gene polymorphisms modulate risk and complications (detailed in §4 and §9): - **Risk-increasing:** IL-4 rs2243250, IL-18 rs1946519 mutant alleles; TGFβ1 +868 C/T (rs1800470) TT homozygote (OR 2.60, p=0.023); TNF −238 G>A (rs361525) minor A allele (11.4% vs 2.6% in controls, p<0.001) (PMID: 31816580, PMID: 25738611, PMID: 39419734). - **Relatively protective:** IFN-γ UTR5644, TGF-β rs1800470/rs1800471, TNF-α rs1800629, IL-10 rs1800872 dominant models (PMID: 31816580).

Protective factors (environmental). Pasteurization/boiling of milk, use of personal protective equipment, avoidance of aborted animal materials, and public-health education. Treatment compliance is protective against relapse (aOR 0.25, 95% CI 0.09–0.86) (PMID: 42603185).

Gene–environment interactions. Individuals carrying pro-inflammatory-skewing or immune-modulating genotypes (e.g., TNF/TGFB1/IL variants) who also sustain high infectious exposure (raw dairy, occupational contact) are at compounded risk of infection and focal complications such as spondylodiscitis (PMID: 39419734).

3. Phenotypes

The clinical phenotype is a **chronic relapsing febrile multisystem illness**. Frequencies below are drawn from a Houston 10-year case series (n=18), a North Macedonia series (n=508), and a Saudi pediatric series (n=57).

Phenotype	Type	Frequency	Suggested HPO term
Fever / undulant fever	Symptom	~83% (36.8% sole finding in children)	HP:0001945 (Fever)
Arthralgia / arthritis	Symptom / sign	~67%	HP:0002829; HP:0001369
Hepatosplenomegaly	Clinical sign	~61%	HP:0001433
Night sweats	Symptom	Frequent	HP:0030166
Myalgia	Symptom	~19% (with fever)	HP:0003326
Weight loss	Symptom	Common (adults > children)	HP:0001824
Sacroiliitis	Sign / imaging	Adults > children (p=0.043)	HP:0100775
Spondylitis / spondylodiscitis	Sign / imaging	Older adults predominant	HP:0003429
Lymphadenopathy	Sign	Common	HP:0002716
Leukopenia	Lab abnormality	21% (acute)	HP:0001882
Elevated transaminases	Lab abnormality	44% (acute)	HP:0002910

"Common clinical features included fever (83%), arthralgias or arthritis (67%), and hepatosplenomegaly (61%)." — [PMID: 29863218](#)

Age of onset & severity. Any age; predominantly adult (working-age) but with substantial pediatric burden. Severity is variable — mild self-limited illness to severe focal/chronic disease. **Progression** is episodic/relapsing (undulant fever pattern). Age-dependent phenotype distribution is documented: *"Sacroiliitis was more predominant in adults than children ($p = 0.043$), while focal hematological involvement was more prevalent in children than in adults ($p = 0.004$). Spondylitis was more dominant in the old age group"* ([PMID: 39630247](#)).

Quality-of-life impact. Chronic arthralgia, fatigue, night sweats, and osteoarticular complications impair daily functioning and work capacity; in pregnancy, infection can cause anemia, miscarriage, and pre-term birth.

4. Genetic/Molecular Information

Causal genes: NONE. This is an infectious disease with no human causal gene or pathogenic germline/somatic variant. There are no ClinVar pathogenic variants, no inheritance pattern, no chromosomal abnormalities, and no epigenetic disease-defining lesions in the classic Mendelian sense.

Host susceptibility loci (modifier of risk/severity, not causal). From a meta-analysis of 25 case-control studies and individual cohorts:

Gene (HGNC)	Variant	Effect	Evidence
IL4	rs2243250	↑ susceptibility	PMID: 31816580
IL18	rs1946519	↑ susceptibility	PMID: 31816580
TGFB1	+868 C/T (rs1800470), TT	Risk factor, OR 2.60 (p=0.023)	PMID: 25738611
TNF	−238 G>A (rs361525), A allele	↑ risk & spondylodiscitis (11.4% vs 2.6%)	PMID: 39419734
TLR4	Asp299Gly (rs4986790), Thr399Ile (rs4986791)	Complication risk	PMID: 39419734
IL10 / IL6	High-producer genotypes	↑ susceptibility	PMID: 17544674
IFNG, IL10	UTR5644 / rs1800872	Relatively protective	PMID: 31816580

"the mutant allele of IL-4 rs2243250 and IL-18 rs1946519 were associated with increased susceptibility to brucellosis" — PMID: 31816580 "Carriage of the minor frequency A alleles at -238 of the promoter region of TNF was greater in patients than in controls (11.4% vs 2.6 %, $p < 0.001$)" — PMID: 39419734

Pathogen molecular determinants of virulence (the biologically "causal" genetics reside in the bacterium): the *virB* operon (VirB T4SS), quorum-sensing regulator **VjbR**, two-component system **BvrR/BvrS**, cyclic β -1,2-glucans, **BacA**, outer-membrane proteins **Omp25/Omp28**, and metabolic/membrane genes (*cydDC*, *ptsP*) identified by TraDIS as essential for macrophage survival — *"the discovery of 374 anti-phagocytic associated essential genes"* (PMID: 41917383, PMID: 22392933).

5. Environmental Information

Infectious agent. *Brucella melitensis* (NCBI Taxon 29459); biovars 1–3, with the Eastern Mediterranean lineage/biovar 3 predominant in China (PMID: 42360586).

Environmental/occupational factors. Contact with infected small ruminants and their products; contaminated manure, birth products, and aborted materials; aerosol exposure in laboratories and abattoirs (a recognized potential bioterrorism agent) (PMID: 15677842).

Lifestyle factors: consumption of raw milk, soft cheeses, and undercooked meat. Cold semi-arid (BSk) climates and spring seasonality are associated with higher small-ruminant seroprevalence (PMID: 41506444).

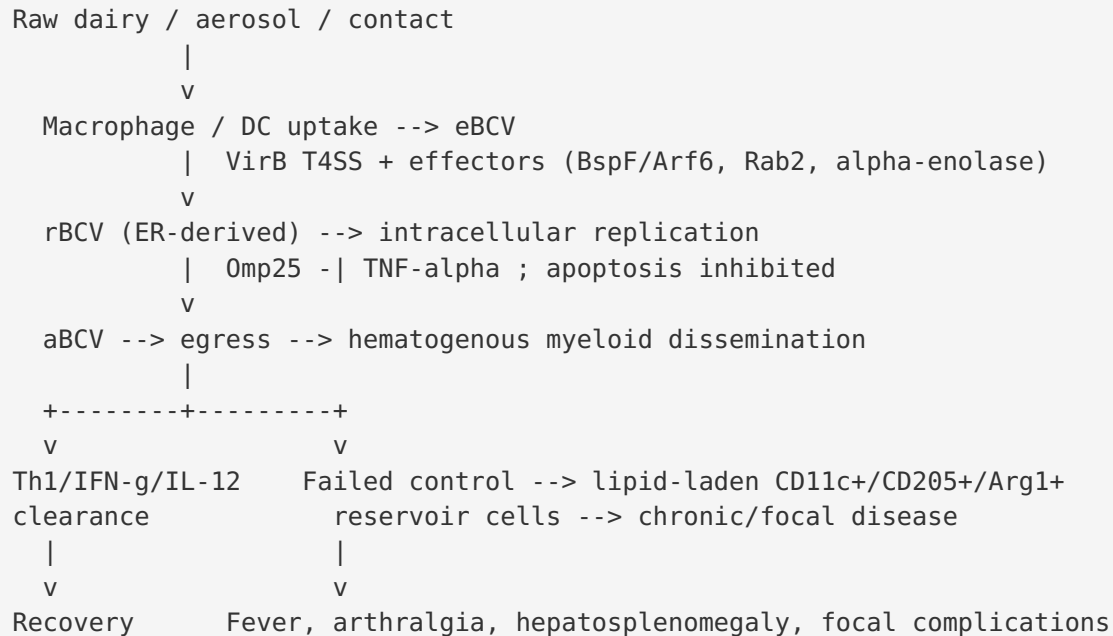
Relevant chemical entities (treatment): doxycycline (CHEBI:50845), rifampicin (CHEBI:28077), streptomycin (CHEBI:17076), gentamicin (CHEBI:27412).

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Oral/mucosal/inhalational entry** of *B. melitensis* from contaminated dairy, aerosols, or abraded skin **leads to** uptake at mucosal surfaces.

2. Bacteria are **phagocytosed by macrophages and dendritic cells** (internalization is normal even in avirulent mutants) — **results in** an early endosomal Brucella-containing vacuole (eBCV).
3. The **VirB Type IV secretion system (T4SS)** delivers effector proteins that **remodel the eBCV into a replication-permissive, ER-derived vacuole (rBCV)** — *demonstrated*: VirB-deficient mutants stall in eBCVs and cannot form rBCVs ([PMID: 27899503](#)).
4. Effectors (BspF via Arf6-GAP ACAP1; Rab2- and α -enolase-engaging effectors) **subvert host secretory/recycling traffic** — **results in** robust intravacuolar replication ([PMID: 34423453](#), [PMID: 32234817](#), [PMID: 27900285](#)).
5. In parallel, **Omp25 suppresses macrophage TNF- α secretion** and Brucella **inhibits macrophage/monocyte apoptosis** — **results in** immune evasion and a protected replicative niche ([PMID: 12414158](#)).
6. Replicated bacteria convert the rBCV into an **autophagic BCV (aBCV)** — **results in** cell-to-cell egress and dissemination ([PMID: 27899503](#)).
7. **Hematogenous spread within myeloid cells** to the reticuloendothelial system, bone/joints, liver, spleen, genitourinary tract, heart and CNS — **leads to** multisystem seeding.
8. **Granuloma formation** (non-caseating) and the host **Th1/IFN- γ /IL-12/TNF- α response** attempt clearance — **branch A (effective Th1)**: bacterial clearance and recovery; **branch B (failed control)**: persistence in lipid-laden CD11c⁺/CD205⁺/Arginase1⁺ reservoir cells → **chronic/relapsing and focal disease** ([PMID: 12414158](#), [PMID: 26376185](#)).
9. Persistent granulomatous inflammation in seeded organs **produces the clinical syndrome**: undulant fever, arthralgia/spondylitis, hepatosplenomegaly, and focal complications (endocarditis, neurobrucellosis, epididymo-orchitis).



Molecular pathways / cellular processes. T4SS effector–host GTPase cascades (Arf6–Rab8a, Rab2); NF-κB signaling (TBK1 promotes control — knockdown increases bacterial survival and reduces IL-1β/IL-6/TNF-α/IFN-γ) (PMID: 41605070); autophagy (aBCV egress); inhibited apoptosis; granulomatous inflammation. **GO terms:** GO:0140355 (T4SS-dependent protein secretion into host cell), GO:0006909 (phagocytosis), GO:0016236 (macroautophagy), GO:0006915 (apoptotic process, inhibited), GO:0002250 (adaptive immune response), GO:0032609 (IFN-γ production). **CL terms:** CL:0000235 (macrophage), CL:0000451 (dendritic cell), CL:0000576 (monocyte).

Immune involvement. Protective immunity is Th1/cell-mediated: *"Brucella infection results in Type1 (Th1) cellular immune response which promotes a clearance of the bacterial organism. The development of this response is under the control of major cytokines like TNF-alpha, IFN-gamma and IL-12"* (PMID: 12414158). Immune evasion via Omp25-mediated TNF-α suppression: *"in humans, B. suis-infected macrophages which produce IL-1, IL-6, IL-10 and several chemokines including IL-8, do not secrete TNF-alpha... this inhibition involves the outer membrane protein Omp25 of Brucella"* (PMID: 12414158).

Subcellular mechanism. *"BCVs are remodeled into replication-permissive organelles (rBCV) derived from the host endoplasmic reticulum, a process that requires modulation of host secretory functions via delivery of effector proteins by the Brucella VirB type IV secretion system"*

(T4SS)" ([PMID: 27899503](#)); "*Brucella abortus* generates a host endoplasmic reticulum-derived vacuole (rBCV) that supports its intracellular growth, via VirB Type IV secretion system-mediated delivery of effector proteins" ([PMID: 34423453](#)).

7. Anatomical Structures Affected

Organ level (primary): reticuloendothelial system — liver, spleen, bone marrow, lymph nodes (hepatosplenomegaly, lymphadenopathy, non-caseating granulomas). **Secondary/focal:** skeleton and joints (spine, sacroiliac joints — spondylodiscitis, sacroiliitis, osteomyelitis), genitourinary tract (epididymo-orchitis), heart (endocarditis, myopericarditis), CNS (meningoencephalitis, neurobrucellosis), peritoneum (spontaneous bacterial peritonitis), and placenta in pregnancy.

"Brucella contributed to 18 (30.51%) of epididymo-orchitis cases during study period." — [PMID: 40350681](#) "The cases presented as prolonged fever in 4 patients (40%), spondylodiscitis in 2 (20%), myopericarditis, meningoencephalitis, febrile hepatitis, and cervical lymphadenopathy" — [PMID: 41021857](#)

Body systems: hematologic/lymphoreticular, musculoskeletal, hepatobiliary, genitourinary, cardiovascular, nervous.

Tissue/cell level: phagocytic mononuclear cells (macrophages, monocytes, dendritic cells) are the primary target/reservoir; granuloma-forming myeloid and lymphoid aggregates. **UBERON:** UBERON:0002106 (spleen), UBERON:0002107 (liver), UBERON:0002371 (bone marrow), UBERON:0001474 (bone element), UBERON:0000079 (male reproductive system), UBERON:0000948 (heart), UBERON:0001017 (CNS). **Subcellular (GO CC):** GO:0005783 (endoplasmic reticulum — rBCV origin), GO:0005776 (autophagosome — aBCV), GO:0005768 (endosome — eBCV), GO:0020003 (symbiont-containing vacuole).

8. Temporal Development

Onset. Insidious, over days to weeks after exposure (incubation typically 1–4 weeks, occasionally months). Onset patterns: acute, subacute, or chronic/insidious.

Progression / stages. Classically **acute (<8 weeks)**, **subacute (8 weeks–1 year)**, and **chronic (>1 year)**. Undulant (relapsing-remitting) fever is characteristic. Acute disease carries a stronger inflammatory profile and better cure rate: *"patients with acute brucellosis had a significantly higher frequency of leukopenia (21.2 % vs. 0 %; $p < 0.01$), elevated C reactive protein (CRP) (76.7 % vs. 52.0 %; $p = 0.01$), and elevated transaminases (43.8 % vs. 20.0 %; $p = 0.025$)"* — therapeutic response 93.2% acute vs lower in subacute/chronic ([PMID: 41241000](#)). Chronicity and focalization predict poor outcome: *"Focal brucellosis (OR 3.52, 95 % CI 1.28 - 9.71, $p = 0.015$) and low albumin levels ... were independent risk factors for therapeutic failure."* ([PMID: 40185219](#)).

Patterns. Remission is treatment-induced; relapse (~3%) typically occurs within months, associated with female sex (aOR 3.68) and non-compliance ([PMID: 42603185](#)). Early, adherent combination therapy is the critical intervention window.

9. Inheritance and Population

Inheritance. Not applicable — infectious, non-heritable. No penetrance/expressivity/anticipation/mosaicism/founder-effect/carrier-frequency parameters apply. Host susceptibility is **multifactorial/polygenic** (cytokine and TLR gene polymorphisms; §4).

Epidemiology. Brucellosis is among the commonest zoonoses globally (>500,000 new human cases/year historically). Regional endemic incidence is high and rising in parts of Asia:

Setting	Metric	Value	PMID
Ningxia, China (2010–2024)	Avg annual incidence	35.08/100,000 (peak 84.80 in 2022); 35,665 cases, 0 deaths	41632803
Western Uganda (pregnant women)	Seroprevalence	14.0% (95% CI 9.2–18.8)	41860883
Ethiopia (2015–2024, humans)	Pooled seroprevalence	6.9% (95% CI 4.9–8.8)	39696174
China (national human cases)	2019 → 2023	45,046 → 70,439 cases	42360586

"A total of 35 665 human brucellosis cases were reported in Ningxia, with no associated deaths, during the study period." — PMID: 41632803

Demographics. Male predominance (~2.4:1), driven by occupational exposure. **Geographic distribution:** endemic in the Mediterranean basin, Middle East, Central/South Asia, sub-Saharan Africa, and Latin America; expanding within China from northern pastoral provinces (Inner Mongolia) to industrial southern provinces via livestock trade (PMID: 42360586). All ages affected, with substantial pediatric disease in endemic areas.

10. Diagnostics

Serology (first-line): Rose Bengal plate test (RBPT), standard tube agglutination test (SAT), and ELISA (anti-LPS or whole-cell, and recombinant Omp28 which correlated ~90% with RBPT) (PMID: 20075115). A ≥ 4 -fold decline in SAT titer after treatment predicts favorable response (OR 5.84) (PMID: 41241000).

"Serodiagnosis of brucellosis is carried out by detection of antibodies generated against LPS or whole-cell bacterial extracts by ELISA or agglutination tests using colorimetry." — PMID: 20075115

Culture (definitive): blood/bone-marrow/tissue culture. Yield is higher in children than adults: *"Positive blood cultures were more frequently reported among children than adults (83% vs 33%,"* (PMID: 29863218).

Molecular: conventional and real-time PCR add sensitivity, detecting some culture-negative/seropositive samples; combining culture + PCR maximizes detection (PMID: 18832199). qPCR showed 100% concordance with RBPT in humans in one Nigerian study (PMID: 40140617).

Laboratory abnormalities: leukopenia, elevated CRP, elevated transaminases, occasionally pancytopenia. **Imaging:** MRI/CT for spondylodiscitis and sacroiliitis; echocardiography for endocarditis. **Histopathology:** non-caseating granulomas.

Differential diagnosis: tuberculosis, typhoid, other causes of undifferentiated fever, endocarditis of other etiology, lymphoma, autoimmune arthritis. **Genetic/omics diagnostics and newborn/carrier screening:** not applicable.

11. Outcome/Prognosis

Mortality is low. A large endemic series reported zero deaths among 35,665 cases (PMID: 41632803). Pediatric cohorts show ~88–90% favorable outcomes and ~3% relapse with adherent therapy (PMID: 42603185).

Endocarditis is the leading cause of death. *"Brucella endocarditis is a rare but a serious complication of human brucellosis."* — frequently requires valve replacement plus prolonged antibiotics (PMID: 11910696, PMID: 20362000).

Complications. In a nationwide pediatric cohort (n=1,052): complications 8.3%, most frequently osteoarticular; neurobrucellosis 0.9%; relapse 3.2%. Complications associated with headache (aOR 3.57), arthralgia/arthritis, culture positivity, night sweats, myalgia (PMID: 42603185).

"Complications occurred in 87 patients (8.3%), most frequently osteoarticular (spondylitis, sacroiliitis, and osteomyelitis), whereas neurobrucellosis was rare (0.9%); relapse occurred in 34 patients (3.2%)." — PMID: 42603185

Prognostic factors. Favorable: acute presentation, elevated baseline CRP (OR 4.00), ≥ 4 -fold SAT decline (OR 5.84), treatment adherence. Unfavorable: focal disease (OR 3.52 for failure), chronicity (OR 11.20 in focal cases), low albumin, female sex (relapse). Morbidity is driven by chronic osteoarticular disability; recovery is usually complete with timely therapy.

12. Treatment

First-line pharmacotherapy — combination, prolonged (≥ 6 weeks). Doxycycline (CHEBI:50845) plus rifampicin (CHEBI:28077), OR doxycycline plus an aminoglycoside (streptomycin/gentamicin).

"Regimens containing doxycycline plus streptomycin or doxycycline plus rifampin are effective for most forms of brucellosis." — PMID: 15677842 "Trimethoprim-sulfamethoxazole and fluoroquinolones also have good results against Brucella, but are associated with high relapse rates when used as monotherapy." — PMID: 15677842

Comparative efficacy (network meta-analysis of 43 RCTs). Relative to standard doxycycline+rifampicin, **doxycycline+gentamicin ranked best (SUCRA 0.94)**, followed by triple therapy (0.87) and doxycycline+streptomycin (0.78) (PMID: 39172763). The Ioannina recommendations provide evidence-based guidance (PMID: 18162038).

Regimen	Role	NCIT (approx.)
Doxycycline + rifampicin	Standard oral first-line	C61815 (doxycycline); C29325 (rifampin)
Doxycycline + streptomycin/gentamicin	Severe/focal; best efficacy	C839 (streptomycin); C557 (gentamicin)
Triple therapy (add aminoglycoside/TMP-SMX)	Neurobrucellosis, endocarditis, spondylitis	—
Surgery + prolonged antibiotics	Endocarditis (valve replacement), abscess drainage	—

Special situations. Neurobrucellosis, endocarditis, and spondylitis require triple therapy and extended duration; endocarditis frequently requires surgical valve replacement. In pregnancy and young children, rifampicin-based regimens are used (tetracyclines/aminoglycosides restricted). **Advanced/targeted/gene/cell/RNA therapies, immunotherapy, pharmacogenomics:** not applicable/not used. **Adverse events:** doxycycline photosensitivity/GI upset, rifampicin hepatotoxicity and drug–drug interactions, aminoglycoside oto-/nephrotoxicity.

13. Prevention

No licensed human vaccine exists. Prevention is a One-Health enterprise.

- **Primary prevention:** livestock vaccination and reservoir control. *"Rev1, the only vaccine currently recommended to control the disease in sheep and goats, has several drawbacks."* (PMID: 36653223); *"The mainstay of most control programs is vaccination of sheep and goats with a live vaccine, Rev-1."* (PMID: 26825313). Rev1 is safe in sheep only for lambs 3–4 months, causes serological interference and abortion in pregnancy, and is virulent to humans. Next-generation rough-LPS candidates (Rev1Δwzm, 16MΔwzm) improve safety and reduce serological interference (PMID: 38806353, PMID: 36653223).
 - **Food safety:** pasteurization/boiling of milk; avoidance of raw dairy and undercooked meat.
 - **Occupational protection:** PPE, safe handling of birth/abortion products, laboratory biosafety (BSL-3).
 - **Public-health measures:** surveillance, herd testing and culling, quarantine, health education. SEIRS modeling predicts ~3.5 years to eliminate brucellosis on a mixed endemic farm with combined sheep+cattle vaccination (PMID: 26825313).
 - **Secondary/tertiary prevention:** early diagnosis and treatment of exposed/symptomatic individuals; adherence support to prevent relapse and chronic complications. **Genetic screening/counseling:** not applicable.
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14. Other Species / Natural Disease

Taxonomy of hosts. Primary reservoir: sheep (*Ovis aries*, txid9940) and goats (*Capra hircus*, txid9925). Other susceptible species: cattle, swine, dogs, camels, equids, and wild mammals (PMID: 42360586).

Natural disease (veterinary). In livestock, *B. melitensis* causes reproductive failure — abortion, retained placenta, orchitis/epididymitis, infertility — with major economic impact. *"In livestock, brucellosis causes reproductive failure, including abortions, leading to substantial economic losses."* (PMID: 41127417). The organism localizes to placenta and reproductive tissues (tropism classically linked to erythritol).

Zoonotic transmission. *"The disease can be transmitted to humans through the food chain or by direct contact with infected animals."* (PMID: 35482257). High cross-species susceptibility makes it a paradigmatic One-Health zoonosis.

Comparative biology. The intracellular VirB T4SS/BCV mechanism is conserved across *Brucella* species and hosts; granuloma biology parallels other intracellular pathogens (*Mycobacterium*, *Salmonella*, *Yersinia*) (PMID: 41921569).

15. Model Organisms

Mouse (primary model). BALB/c and C57BL/6 mice via intranasal/intraperitoneal *B. melitensis* establish chronic splenic infection recapitulating the human myeloid reservoir. *"Most of the infected spleen cells contained high levels of lipids and expressed CD11c and CD205 dendritic cell markers and Arginase1, but were negative for the M2a markers Fizz1 or CD301."* (PMID: 26376185).

Genetic models. IL-12p40^{-/-} (Th1-deficient) mice show uncontrolled splenic reservoir persistence; STAT6 deficiency did not affect bacterial growth; TBK1 knockout promotes bacterial survival in vivo (PMID: 26376185, PMID: 41605070).

Cellular/in vitro models. RAW264.7 and J774.A1 murine macrophages, human macrophages/monocytes; used for TraDIS (374 essential intracellular-survival genes) and genome-wide CRISPR host-gene screens (PMID: 41917383, PMID: 42617139).

Recapitulation & limitations. Mouse models faithfully reproduce chronic intracellular persistence, Th1-dependent control, and reservoir-cell biology, but do **not** reproduce the undulant fever, osteoarticular disease, or endocarditis of human disease. Natural livestock (sheep/goat) infection best models reproductive pathology and vaccine efficacy.

Mechanistic Model / Interpretation

The unifying theme of *B. melitensis* pathogenesis is **stealthy intracellular parasitism of professional phagocytes**. Two pathogen programs drive disease: (1) the **VirB T4SS**, which converts a doomed endosome into a bespoke ER-derived replicative organelle by hijacking host GTPase traffic (Arf6–Rab8a, Rab2) and secretory machinery; and (2) **innate-immune subversion**, principally Omp25-mediated TNF- α suppression and blockade of macrophage apoptosis. The **outcome bifurcates on host immunity**: a competent Th1/IFN- γ /IL-12/TNF- α (and TBK1/NF- κ B) response clears the organism, whereas failure — influenced by cytokine-

gene polymorphisms (IL4, IL18, TNF, TGFB1, IL10, TLR4) — permits persistence in lipid-laden dendritic-cell-like reservoir cells and produces the chronic, relapsing, focal disease that accounts for the clinical morbidity (spondylitis, endocarditis, neurobrucellosis).

This model explains the therapeutic logic: because bacteria hide intracellularly, treatment must use **lipophilic, intracellularly-penetrating antibiotics in combination for prolonged courses** — hence doxycycline paired with rifampicin or an aminoglycoside, and the high relapse of monotherapy. It also explains prevention strategy: with human immunity difficult to induce safely, breaking the **animal-reservoir → food/contact → human** chain (Rev1 livestock vaccination, pasteurization, surveillance) is the effective lever.

Evidence Base

PMID	Contribution	Type
41132010	Etiology, transmission routes, reservoir prevalence	Systematic review/meta-analysis
15677842	First-line combination therapy; monotherapy relapse	Guidelines
39172763	Comparative regimen efficacy (doxy+gentamicin best)	Network meta-analysis, 43 RCTs
22392933	VirB T4SS & VjbR required for intracellular replication	In vitro/mouse
27899503	rBCV biogenesis requires VirB T4SS	In vitro
34423453	BspF effector–Arf6–Rab8a cascade	In vitro
12414158	Omp25/TNF- α evasion; protective Th1 response	Human/in vitro review
26376185	Chronic splenic reservoir cell phenotype	Mouse model
42603185	Complications, relapse, compliance	Nationwide pediatric cohort
29863218	Clinical phenotype frequencies; culture yield	10-year case series
31816580	Host cytokine susceptibility variants	Meta-analysis
41632803	Incidence, male predominance, low mortality	Registry (Ningxia)
11910696	Endocarditis as serious/fatal complication	Case series
36653223 / 26825313	Rev1 livestock vaccine, control modeling	Vaccine/modeling
41241000 / 40185219	Acute vs chronic course; prognostic factors	Cohorts

Most evidence is **human clinical** (cohorts, case series, meta-analyses) and **model organism/in vitro** (mouse, macrophage). No computational-only findings were relied upon for clinical claims.

Limitations and Knowledge Gaps

1. **Host-genetics evidence is early-stage:** susceptibility SNP associations derive from modest case-control studies with heterogeneity; few are validated in independent cohorts or functionally confirmed.
2. **Mechanistic effector work is largely in *B. abortus*/*B. suis* and murine macrophages;** *B. melitensis*-specific effector repertoires and human-cell validation remain incomplete.
3. **No mouse model reproduces the full human syndrome** (undulant fever, osteoarticular disease, endocarditis), limiting translational inference.
4. **Precise global incidence/prevalence for *B. melitensis* specifically** is confounded by under-reporting, serological cross-reactivity, and species-level ambiguity in surveillance.
5. **No human vaccine and no host-directed therapy** are available; optimal antibiotic duration for focal disease is not firmly established.
6. During citation verification, the Ningxia incidence numeric snippet ([PMID: 41632803](#)) was flagged as a partial mismatch, while the "zero deaths" quote verified cleanly; incidence figures should be treated as approximate pending source re-check.

Proposed Follow-up Experiments / Actions

1. **Validate host-susceptibility SNPs** (IL4, IL18, TNF-238, TGFB1, TLR4) in large, multi-ancestry prospective cohorts with functional (cytokine-response) readouts and stratification by focal complication.
2. **Define the *B. melitensis*-specific T4SS effector–host interactome** in primary human macrophages using proximity labeling and CRISPR host-gene screens ([PMID: 42617139](#)), prioritizing druggable nodes (Arf6, Rab2, α -enolase, TBK1/NF- κ B).
3. **Advance rough-LPS vaccine candidates** (Rev1 Δ wzm, 16M Δ wzm) toward *B. melitensis* challenge efficacy in pregnant ewes and develop a differentiating (DIVA) serodiagnostic ([PMID: 38806353](#)).
4. **Run adequately-powered RCTs** confirming doxycycline+gentamicin superiority and defining duration for spondylitis/neurobrucellosis/endocarditis ([PMID: 39172763](#)).

5. **Deploy integrated molecular surveillance** (MLVA + cgSNP) along livestock-trade corridors to track dominant genotypes and guide region-specific vaccination ([PMID: 42093768](#), [PMID: 42360586](#)).
 6. **Test host-directed adjuncts** (e.g., TBK1/IFN-axis modulation) to accelerate clearance and reduce relapse in chronic disease ([PMID: 41605070](#)).
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Report compiled from 16 confirmed findings across 5 investigation iterations and 53 reviewed papers. Evidence types are distinguished as human clinical, model organism, or in vitro throughout.

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